Pollard rho across the strata of an isogeny volcano

Question. Take one ECDLP instance, carry it by explicit isogenies to every level of an ℓ-isogeny volcano, and run Pollard rho on each curve: does the level (crater → floor), or the particular curve, change how fast rho solves it? Report min, max and mean.

Answer (measured on two curves: A, `run-20261016T2302Z`, ramified crater; B, `run-20261016T2315Z`, split crater). No measurable effect on either. Curve A in detail: All 29 isogenous curves and 3 isomorphic controls took the same work to within sampling noise. Pooled over the 2,900 isogenous solves: min 8,477, mean 659,263 ± 12,422 (95 % CI), max 2,099,707 group operations, against $\sqrt{(\pi n/2)}$ = 654,547. Every level's 95 % CI includes or touches that value, and per-step cost is flat (641–661 ns/op on every curve). That is what theory predicts: isogenous curves over F_p have the same group order, and generic rho depends only on the order of ⟨P⟩. The spread *between* curves (per-curve means 572,846 – 733,141) matches the spread of 100-trial means of a quantity whose single-solve sd is about 341,000 (sd of a mean ≈ 34,000).

Method

`volcano-rho` (`src/bin/volcano_rho.rs`, native Rust, all statistics computed there; `plot.py` only lays out recorded values):

1. **Curve search.** Random curves $y^2 = x^3 + ax + b$ over random 40-bit primes $p \equiv 3 \pmod 4$, point count by baby-step giant-step, until $D = t^2 - 4p = f^2 \cdot D_K$ has $v_3(f) = 3$ (a 3-volcano with three levels below the crater) and $\#E = h \cdot n$ with n prime, $h \le 64$.
2. **Walk.** Breadth-first over 3-isogenies. Each edge's kernel is a rational root x_0 of the 3-division polynomial $\psi_3 = 3x^4 + 6ax^2 + 12bx - a^2$, and the map is Vélu's, applied explicitly to points. Vertices are keyed by j-invariant. A vertex's level is `depth − dist(v, floor)`, the floor being the vertices with exactly one rational 3-kernel; the tool asserts that no edge skips a level.
3. **One problem everywhere.** P of order n and $Q = kP$ on the root curve are pushed along the BFS tree to every curve. Each edge is checked: images lie on the codomain, $\varphi(P+Q) = \varphi(P) + \varphi(Q)$, and $n \cdot \varphi(P) = O$ with $\varphi(P) \neq O$ ($3 \nmid n$, so every map is injective on ⟨P⟩). Every rho solve on every curve returned the same k.
4. **Rho.** Single walker, 32-adding walk (Teske), distinguished points with $\theta = 2^{-8}$, no negation map; restarts on a DP-free run longer than $40/\theta$. Trial seed s uses the same walk constants on every curve (the partition hashes x, so walks differ). Measured unit: group additions to the solve; also wall time per solve.
5. **Control.** Three random isomorphic re-scalings of the root, (u^4a, u^6b), $(x, y) \mapsto (u^2x, u^3y)$: the same curve in new coordinates, showing the spread that a pure change of representation produces.
6. All 3,200 (arm, seed) jobs shuffled and run on 4 threads.

Curve A: the instance (`run-20261016T2302Z`, ramified crater)

p	1090993606999 (40 bits)
root E	a = 690282649366, b = 1050606678233
#E	1090993806052 = 4 · n, n = 272748451513 (prime, 38 bits)
trace t	−199052
$D = t^2 - 4p$	$-4324352729292 = 54^2 \cdot (-1482974187)$; $v_3(f) = 3$
crater	$D_K \equiv 0 \pmod 3$: ramified, 2 crater vertices joined by one edge
k	49631396114 (Q = kP)
volcano	80 curves = 2 + 6 + 18 + 54 (crater, L1, L2, floor), 79 edges, all verified

Curve A results (iterations = group operations; time on a shared 4-vCPU VM) Pooled by level:

level	curves	solves	min	mean ± 95 % CI	median	max	mean ms ± CI
0 crater	2	200	17,415	700,483 ± 47,849	662,463	1,621,252	456.9 ± 32.0
1	6	600	48,186	681,766 ± 26,525	656,746	1,861,764	440.7 ± 17.6
2	10	1,000	9,649	637,158 ± 20,410	598,424	1,792,889	414.4 ± 13.6
3 floor	11	1,100	8,477	659,589 ± 20,993	620,458	2,099,707	426.0 ± 13.9
all isogenous	29	2,900	8,477	659,263 ± 12,422	622,386	2,099,707	427.1 ± 8.2
isomorphic control	3	300	23,698	674,042 ± 40,816	628,245	1,820,972	434.0 ± 27.0

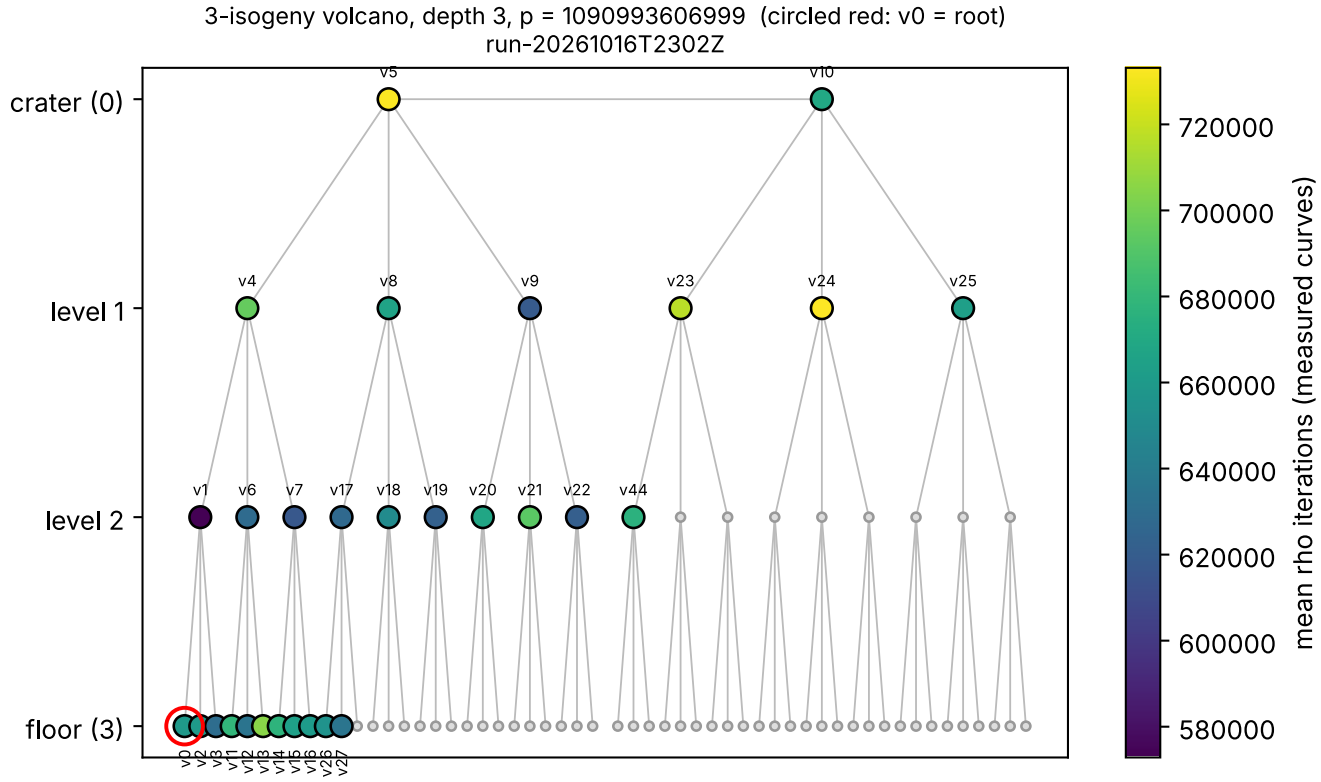
Reference: $\sqrt{(\pi n/2)}$ = 654,547 (+ about 2^8 for the distinguished-point tail).

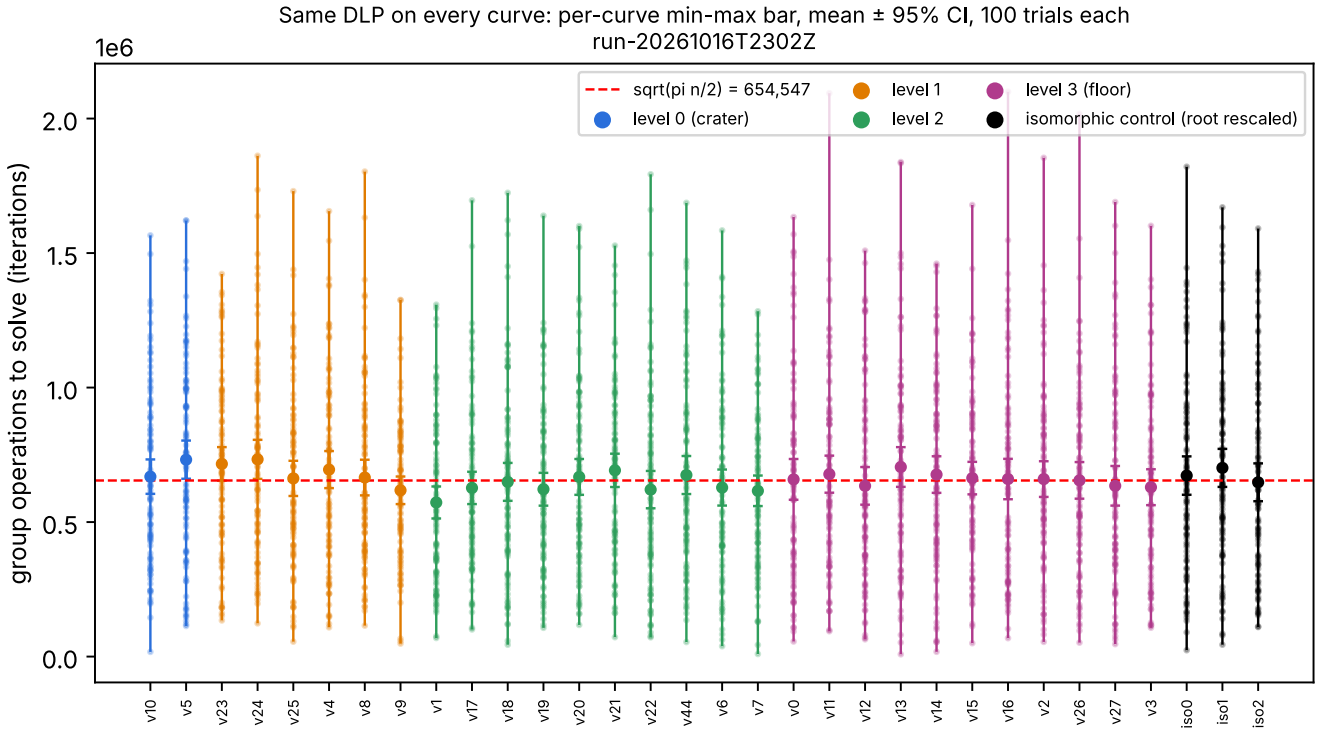
Per curve (100 solves each), iterations:

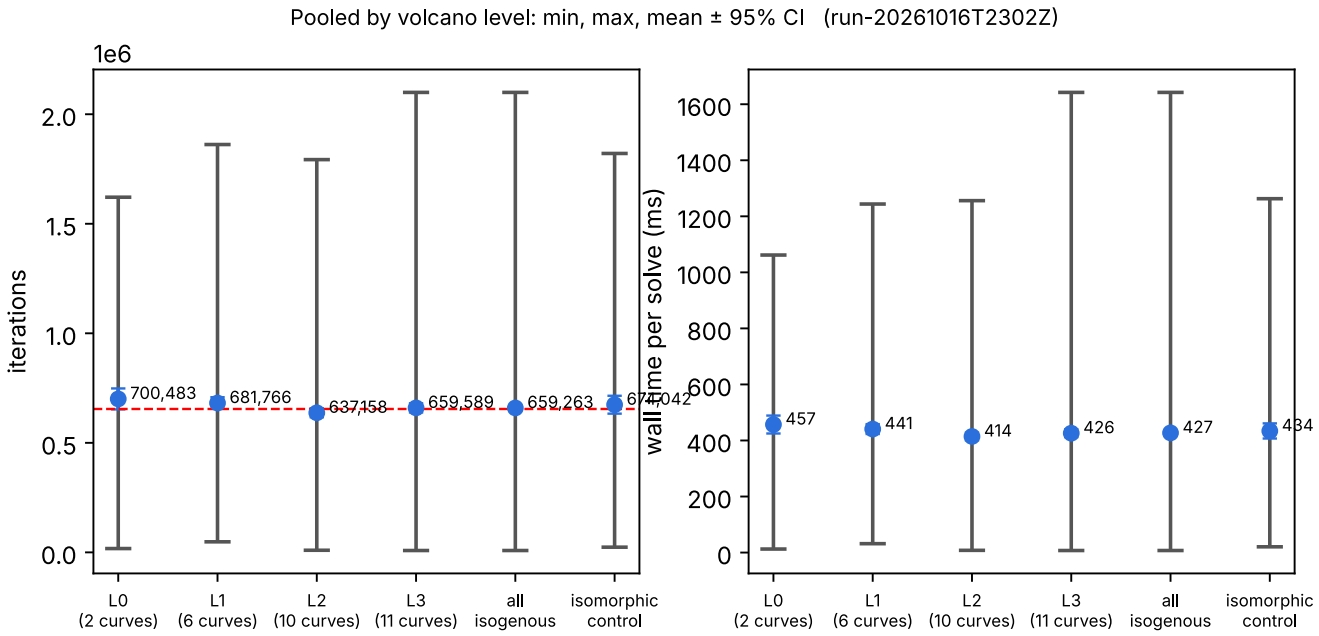
curve	level	min	mean	max		curve	level	min	mean	max
v5	0	113,746	732,044	1,621,252		v18	2	44,056	649,444	1,724,073

v10	0	17,415	668,921	1,565,882		v19	2	106,910	621,995	1,638,981
v4	1	108,640	695,082	1,655,814		v20	2	117,650	667,890	1,600,331
v8	1	115,038	665,553	1,802,980		v21	2	72,048	692,313	1,527,838
v9	1	48,186	618,080	1,325,879		v22	2	71,551	620,647	1,792,889
v23	1	134,281	716,324	1,422,523		v44	2	54,056	674,911	1,686,754
v24	1	123,525	733,141	1,861,764		v0 (root)	3	55,979	658,723	1,634,032
v25	1	55,435	662,413	1,730,218		v2	3	54,631	659,817	1,854,099
v1	2	69,112	572,846	1,308,207		v3	3	107,168	629,575	1,601,367
v6	2	38,512	628,345	1,584,239		v11	3	94,240	678,053	2,094,241
v7	2	9,649	616,240	1,283,603		v12	3	64,211	634,612	1,509,154
v17	2	100,833	626,950	1,696,010		v13	3	8,477	704,738	1,838,374
iso0	ctrl	23,698	672,888	1,820,972		v14	3	17,758	676,654	1,460,561
iso1	ctrl	44,112	701,414	1,670,056		v15	3	49,801	663,488	1,678,869
iso2	ctrl	109,711	647,822	1,591,405		v16	3	68,503	659,946	2,099,707
						v26	3	51,815	655,015	2,017,181
						v27	3	46,289	634,861	1,689,685

Full per-curve rows (with times, ns/op, medians, p10/p90) are in `runs/run-20261016T2302Z/results.json`; every solve is a row of `trials.csv`.







Curve B: a second volcano with a split crater (`run-20261016T2315Z`)

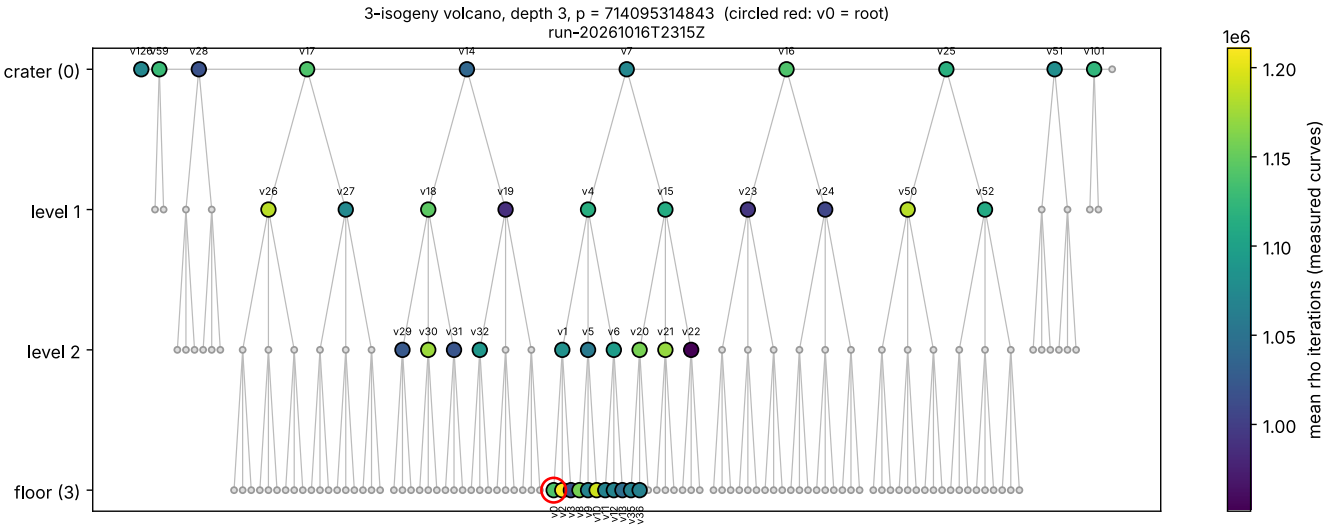
Same tool, `--crater split --seed 20261017`. This volcano's crater is a cycle, so the walk reaches a long run of crater vertices, each with its own depth-3 tree. The cycle is truncated at the walk radius.

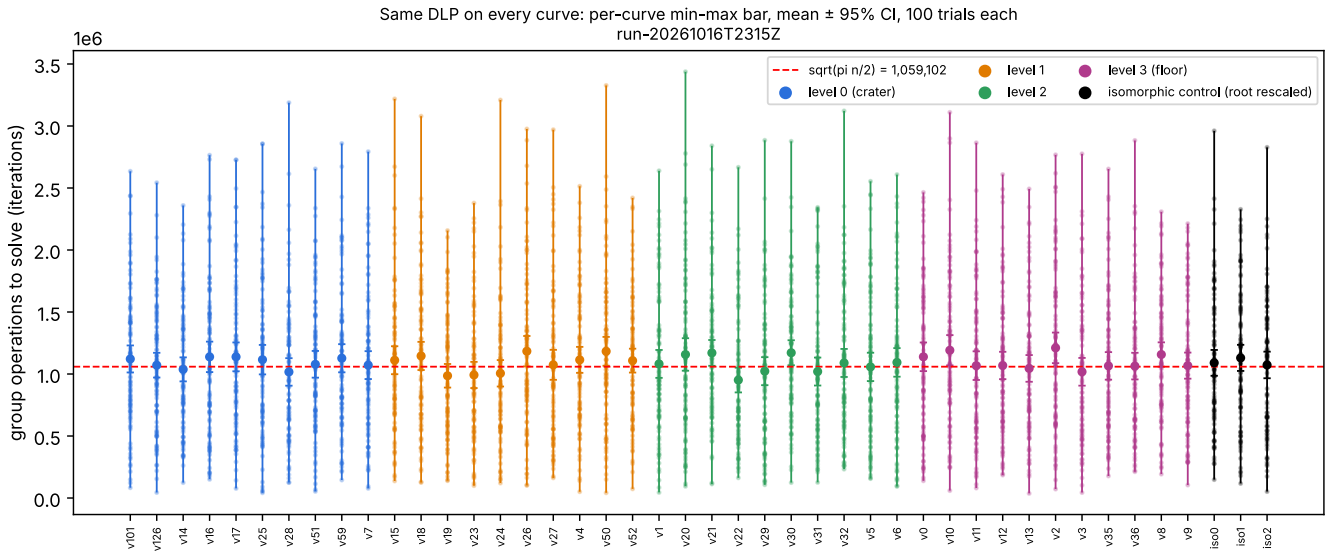
p	714095314843 (40 bits)
root E	a = 354455170800, b = 66680642576
#E	714094928137 = n, prime (cofactor 1)
trace t	386707
D = t² − 4p	−2706838955523 = 27² · (−3713084987); $v_3(f) = 3$
crater	D_K ≡ 1 (mod 3): split, the crater is a cycle
k	661235192333
walk	323 curves, 322 edges verified; inside the measured radius 11 crater, 18, 42 and 90 floor curves by level

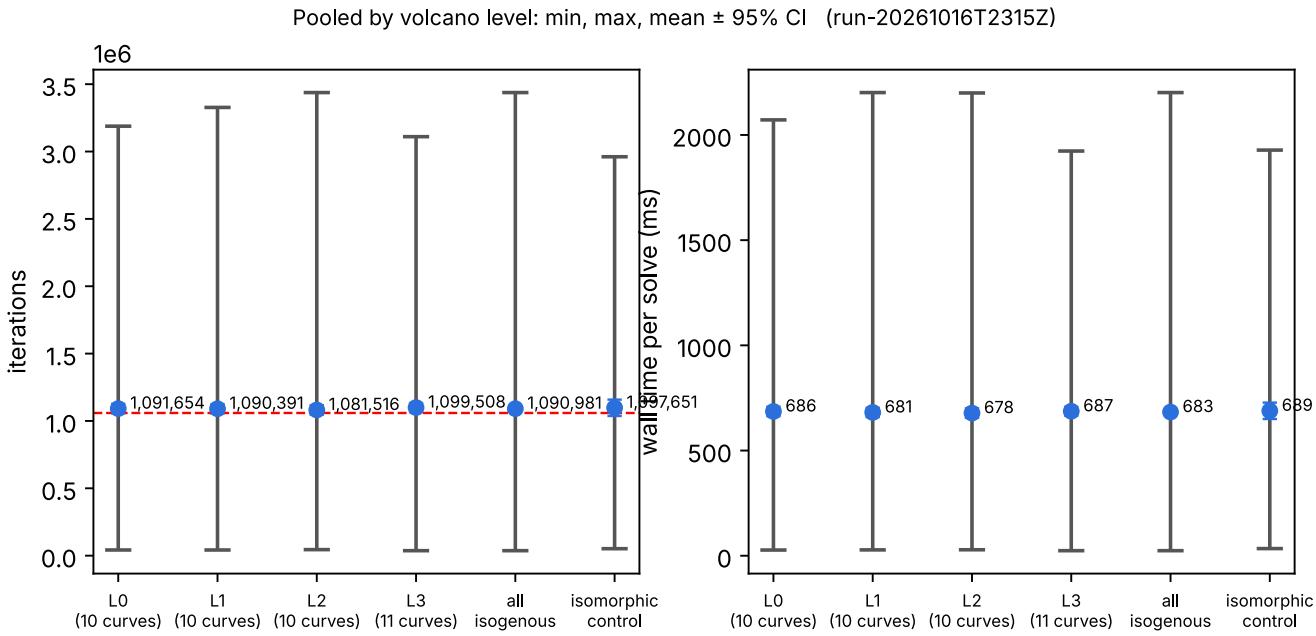
rho	41 curves + 3 controls × 100 solves = 4,400, all correct; 754 s

level	curves	solves	min	mean ± 95 % CI	median	max	mean ms ± CI
0 crater	10	1,000	42,088	1,091,654 ± 35,041	1,032,925	3,188,276	685.7 ± 22.6
1	10	1,000	42,018	1,090,391 ± 34,841	1,046,357	3,327,413	681.2 ± 22.0
2	10	1,000	44,771	1,081,516 ± 35,498	1,012,223	3,437,779	678.1 ± 22.7
3 floor	11	1,100	36,930	1,099,508 ± 33,756	1,050,592	3,110,345	686.8 ± 21.1
all isogenous	41	4,100	36,930	1,090,981 ± 17,380	1,034,403	3,437,779	683.0 ± 11.0
isomorphic control	3	300	51,690	1,097,651 ± 60,787	1,094,657	2,961,196	688.9 ± 38.9

√(πn/2) = 1,059,102. Per-curve means span 951,361 (v22) to 1,210,896 (v2); the per-op cost is 622–635 ns on every curve. Per-curve rows are in `runs/run-20261016T2315Z/results.json`.







Reading it

- **Observed:** level means differ by at most 63,000 iterations (L0 vs L2), and the 95 % CIs of L0 and L2 overlap. The CIs of L0, L2 and L3 contain $\sqrt{(\pi n/2)}$; L1's lower bound (655,241) sits 700 above it, with four level intervals and no correction for multiple comparisons. Single solves range over 250× (8,477 to 2,099,707) on every level and on the control, so min and max are the extremes of a wide distribution and say little about a curve. The mean is the meaningful statistic.
- **Curve B** (observed): the level means lie within 1.7 % of each other (1,081,516 to 1,099,508), and every level's CI overlaps every other's. All of them, and the control, sit about 3 % above $\sqrt{(\pi n/2)}$. The pooled mean, 1,090,981 ± 17,380, excludes it. Curve A's pooled mean sits 0.7 % above (inside its CI). A known heuristic (derived, not measured here) is that an r-adding walk needs about $1/\sqrt{(1 - 1/r)}$ times a truly random walk's steps: ×1.016 for r = 32, or 1,076,000 on B and 665,000 on A. Both pooled CIs contain those values. The offset is a property of the walk, shared by every curve and the control, not of any stratum.
- **Interpretation (derived, not measured):** an ℓ-isogeny with $\ell \nmid n$ is a group isomorphism $\langle P\rangle \longrightarrow \langle \varphi(P)\rangle$, so generic rho sees the same group of order n on every stratum; what changes between levels is the endomorphism ring, which generic rho does not use. The per-curve differences are what re-drawing the walk's random partition gives, as the isomorphic control (same curve, new coordinates) shows. Walking the volcano is not a way to make this generic DLP cheaper.
- **Gap to matched rho:** this *is* rho on every arm; no index-calculus arm is involved, so there is no IC baseline to improve and no §5a band to rank across.

Limits

- One instance per curve, two 40-bit primes, two volcanoes (ℓ = 3, depth 3; one ramified crater with 2 vertices, so A's L0 has only 200 solves; one split crater, truncated at the walk radius).
- Not an ecbench session: the counted unit is this tool's group-addition count; no isolation (shared 4-vCPU VM, 4 threads, jobs interleaved), so wall times are indicative only. Per-op time includes affine inversion by `i128` extended Euclid.
- Curves measured: A 29 of 80 (the whole volcano), B 41 of the 323 walked (root plus up to 10 per level in BFS order).
- No negation map and no endomorphism-accelerated rho; a curve at the crater with a larger endomorphism ring would only help an algorithm that uses that ring (GLV-style speedups exist only for tiny |D_K|, and this D_K = −1482974187 is not tiny).

Run it on your own curve

```
scripts/volcano_ecdlp.sh --curve P,A,B [--order N] [--ell 3] [--seed S] [--trials 100]
scripts/volcano_ecdlp.sh --ca /path/to/ca --curve P,A,B ...   # cryptanalysis's C rho
```

`y² = x³ + Ax + B` over `F_P`, any prime `P < 2^62`. Without `--order` the tool counts points (baby-step giant-step). It checks the order against random points, takes `n` the largest prime factor, draws a generator (cofactor × a seeded random point) and a secret `k` from `--seed`, and reads the volcano depth `v_ℓ(f)` and the crater shape from `D = t² − 4p`. Then it walks, carries `(P, kP)` to every curve and measures. A curve whose conductor ℓ does not divide has depth 0, so the walk covers crater curves only. Outputs go to `--out` (default `volcano-ecdlp-<UTC>/`): `results.json`, `trials.csv`, `instances.jsonl` (every curve's instance in `ca` format, ready for `ca solve`) and the standard graphs.

- **Engines.** `native` is this binary's single-walker 32-adding walk with distinguished points. The DP rate defaults to `min(8, log₂n/2 − 6)` bits, so small groups keep every point. With every point distinguished, the x-keyed table also catches `±X` matches, so expect about $\sqrt{(\pi n/4)}$. `--engine ca` (or `--ca` on the script) runs `ca solve --alg rho` from aburan28/cryptanalysis on each curve, with its defaults (negation map on) plus any `--ca-arg`. Iterations are then `ca`'s reported `ops` and times its reported `seconds`.
- **Cost.** About $\sqrt{n}$ group operations per solve: n ≈ 2^40 takes a second or so per solve on the native engine. Choose `--trials` with that in mind.

- **Determinism.** The same command and seed give the same curves, instance and iteration counts; wall times vary.

Reproduce

```
cargo build --release --bin volcano-rho
./target/release/volcano-rho --bits 40 --ell 3 --depth 3 --seed 20261016 \
    --trials 100 --per-level 10 --controls 3 --out OUT
./target/release/volcano-rho --bits 40 --ell 3 --depth 3 --crater split --seed 20261017 \
    --trials 100 --per-level 10 --controls 3 --out OUT_B
python plot.py OUT     # matplotlib, plotting only
```

The curve, volcano, instance and every solve's iteration count are a deterministic function of `--seed`; wall times are not. Run from the then-uncommitted `src/bin/volcano_rho.rs` on top of `9b09d308e` (committed in this PR with formatting and lint-only changes), Intel Xeon @ 2.80 GHz, 4 threads, 342.5 s for the rho phase.

Curve B ran on the committed binary (with `--crater`), on the same host, in 754.0 s for the rho phase.

Replay. The committed binary (formatting and lint fixes only) re-run with `--trials 5` reproduces 160/160 iteration counts of the first five seeds of every arm exactly.

Report PDF: `report.pdf`.